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Package structure and manufacturing method thereof

Abstract

A semiconductor package and a manufacturing method for the semiconductor package are provided. The semiconductor package includes a redistribution layer, a semiconductor die, conducting connectors, dummy bumps and an underfill. The semiconductor die is disposed on a top surface of the redistribution layer and electrically connected with the redistribution layer. The conducting connectors are disposed between the semiconductor die and the redistribution layer, and are physically and electrically connected with the semiconductor die and the redistribution layer. The dummy bumps are disposed on the top surface of the redistribution layer, beside the conducting connectors and under the semiconductor die. The underfill is disposed between the semiconductor die and the redistribution layer and sandwiched between the dummy bumps and the semiconductor die. The dummy bumps are electrically floating. The dummy bumps are in contact with the underfill without contacting the semiconductor die.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application claims the priority benefit of U.S. provisional application Ser. No. 63/181,128, filed on Apr. 28, 2021. The entirety of the above-mentioned patent application is hereby incorporated by reference herein and made a part of this specification.

BACKGROUND

(1) During the packaging processes of the semiconductor dies, conductive connection structures and metallic routing structures are formed for routing and interconnecting the dies and/or semiconductor devices in the packages.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.
- (2) FIG. 1 to FIG. 7 are schematic cross-sectional views of various stages in a manufacturing method of a semiconductor package according to some exemplary embodiments of the present disclosure.
- (3) FIG. 8 is a schematic cross-sectional view of a package structure with dummy bumps according to an exemplary embodiment of the present disclosure.
- (4) FIG. 9 and FIG. 10 are schematic planar views illustrating the exemplary layout of the dummy bumps in a package structure according to some exemplary embodiments of the present disclosure.
- (5) FIG. 11A to FIG. 11D are schematic cross-sectional views illustrating the bump structures according to some embodiments of the present disclosure.
- (6) FIG. 12A to FIG. 12C are schematic enlarged partial cross-sectional views showing the dummy bump attached to a portion of a package structure according to some exemplary embodiments of the present disclosure.
- (7) FIG. 13 and FIG. 14 are schematic cross-sectional views of package structures with dummy bumps according to some exemplary embodiments of the present disclosure.

DETAILED DESCRIPTION

(8) The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

(9) Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

(10) In addition, terms, such as “first,” “second,” “third,” “fourth,” and the like, may be used herein for ease of description to describe similar or different element(s) or feature(s) as illustrated in the figures, and may be used interchangeably depending on the order of the presence or the contexts of the description.

(11) Other features and processes may also be included. For example, testing structures may be included to aid in the verification testing of the 3D packaging or 3DIC devices. The testing structures may include, for example, test pads formed in a redistribution layer or on a substrate that allows the testing of the 3D packaging or 3DIC, the use of probes and/or probe cards, and the like. The verification testing may be performed on intermediate structures as well as the final structure. Additionally, the structures and methods disclosed herein may be used in conjunction with testing methodologies that incorporate intermediate verification of known good dies to increase the yield and decrease costs.

(12) FIG. 1 to FIG. 7 are schematic sectional views of various stages in a method of fabricating a package structure in accordance with some embodiments of the present disclosure. FIG. 9 and FIG. 10 are schematic planar views illustrating the exemplary layout of the dummy bumps in a package structure according to some exemplary embodiments of the present disclosure.

(13) Referring to FIG. 1, a carrier **102** with a debonding layer **104** coated thereon is provided and a redistribution layer **110** is formed over the carrier **102**. In some embodiments, the carrier **102** includes any suitable semiconductor carrier for the manufacturing method of the integrated fan-out (InFO) package structure. In some embodiments, the carrier **102** is a glass carrier or a temporary carrier. In some embodiments, the debonding layer **104** is formed from any material suitable for bonding and debonding the carrier **102** from the above layer(s) or any wafer(s) disposed thereon. In some embodiments, the debonding layer **104** includes a light-to-heat-conversion (LTHC) release coating film, possible for room temperature debonding from the carrier **102** by applying laser irradiation. In some embodiments, the debonding layer **104** includes an ultra-violet (UV) glue, which loses its adhesive property when exposed to UV lights. In certain embodiments, the debonding layer **104** may be dispensed as a liquid and cured, or may be a laminate film laminated onto the carrier **102**.

(14) Referring to FIG. 1, once the carrier **102** and the debonding layer **104** are provided, a redistribution layer **110** is formed on the debonding layer **104** over the carrier **102**. In some embodiments, the redistribution layer **110** includes dielectric layers **106a**, **106b**, **106c**, **106d**, **106e** and conductive layers **108a**, **108b**, **108c**, **108d**, **108e** arranged in alternation, and the conductive

layers **108a**, **108b**, **108c**, **108d**, **108e** are sandwiched between the dielectric layers **106a**, **106b**, **106c**, **106d**, **106e**. For example, the conductive layer **108a** may be constituted by a plurality of metallic redistribution patterns. In some embodiments, the formation of the redistribution layer **110** includes forming a layer of a dielectric material (not shown), patterning the layer of dielectric material to form openings, depositing a metallic material filling up the openings to form metallization patterns. Depending on the number of the layers to be formed, these processes may be repeated several times, and the sequentially layers may be denoted based on the formation sequence. In some embodiments, the dielectric layers **106a**, **106b**, **106c**, **106d**, **106e** further include via openings revealing the underlying layer. In some embodiments, the conductive layers **108a**, **108b**, **108c**, **108d**, **108e** may include electrically connected routing traces or fan-out traces, some of them are interconnected with one another by the conductive vias. In some embodiments, the topmost conductive layer **108e** may include bond pads **P1**. In some embodiments, the topmost conductive layer **108e** also includes other routing traces for interconnecting the bond pads **P1**. In some embodiments, the bond pads **P1** may include under bump metallurgy (UBM) portions.

(15) For simplicity, the dielectric layers **106a**, **106b**, **106c**, **106d**, **106e** may be regarded as one single dielectric structure and the conductive layers **108a**, **108b**, **108c**, **108d**, **108e** may be illustrated as conductive redistribution patterns embedded in the dielectric layers. However, from the perspectives of the manufacturing process, the dielectric layers **106a**, **106b**, **106c**, **106d**, **106e** are formed in sequence as five dielectric layers, and the redistribution conductive layers **108a**, **108b**, **108c**, **108d**, **108e** are also formed in sequence and each sandwiched between the two adjacent dielectric layers.

(16) In some embodiments, the materials of the conductive layers **108a**, **108b**, **108c**, **108d**, **108e** include aluminum, titanium, copper, nickel, tungsten, cobalt and/or alloys thereof. The conductive layer **108a**, **108b**, **108c**, **108d**, or **108e** may be formed by, for example, electroplating, deposition, and/or photolithography and etching. In some embodiments, the conductive layer **108a**, **108b**, **108c**, **108d**, or **108e** may further optionally includes a seed layer if formed by plating. In some embodiments, the materials of the dielectric layers **106a**, **106b**, **106c**, **106d**, **106e** include polyimide, epoxy resin, acrylic resin, phenol resin, benzocyclobutene (BCB), polybenzooxazole (PBO), or any other suitable polymer-based dielectric material. The dielectric layer **106a**, **106b**, **106c**, **106d**, or **106e**, for example, may be formed by suitable fabrication techniques such as spin-on coating, chemical vapor deposition (CVD), plasma-enhanced chemical vapor deposition (PECVD), or the like.

(17) It is noted that the number of the conductive layers and the number of the dielectric layers of the redistribution layer **110** illustrated in FIG. 1 are merely for illustrative purposes, and the disclosure is not limited thereto. In some alternative embodiments, less or more layers of the conductive layers and/or dielectric layers may be formed depending on the design requirement.

(18) As illustrated in FIG. 2, corresponding to the subsequently mounted semiconductor dies and other later performed processes to the structure, at least one die attaching region DA and at least one non-die attaching region NDA are predetermined and defined. Also, the scribe lanes SL are predetermined and are located within the non-die attaching region(s) NDA.

(19) As illustrated in FIG. 2, after forming the redistribution layer **110**, bonding portions **112** are respectively formed on the bond pads **P1** of the topmost conductive layer **108e**. In one embodiment, some of the bonding portions **112** are formed on the redistribution layer **110** (on the topmost dielectric layer) but not on the bond pads **P1**. In some embodiments, bonding portions **112** includes conducting bumps **113** located within the die attaching region DA and dummy bumps **114** distributed over the die attaching region DA and the non-die attaching region NDA. In embodiments, the conducting bumps **113** function as active bumps for electrically connecting the redistribution layer **110** with other active semiconductor components or devices. In one embodiment, some of the dummy bumps **114** are not located on the bond pads **P1** and are formed on the redistribution layer **110** (on the topmost dielectric layer). In FIG. 2, it is seen that the

distribution span DD of the dummy bumps **114** is partially overlapped with the die attaching region DA.

(20) In some embodiments, the conducting bumps **113** include micro-bumps, metallic posts such as copper posts or copper alloy posts, metallic posts with solder materials/solder pastes, or metallic posts with suitable metallic or metal alloy coatings for enhancing bonding, or other suitable metallic connectors. In some embodiments, the dummy bumps **114** include micro-bumps, metallic posts such as copper posts or copper alloy posts, metallic posts with solder materials/solder pastes, or metallic posts with suitable metallic or metal alloy coatings for enhancing bonding, or other suitable metallic connectors. In some embodiments, the bonding portions **112** may be formed through electroplating. In some embodiments, the conducting bumps **113** and the dummy bumps **114** of the bonding portions **112** are formed at the same time through the same electroplating process. In some embodiments, formation of the bonding portions **112** involves forming a seed layer (not shown) by sputtering, forming a mask (not shown), forming metallic posts by electroplating and then etching off the extra seed layer and the mask. In some embodiments, formation of the bonding portions **112** further involves forming a metallic coating on the metallic posts and/or forming solder caps on the metallic posts by printing or plating.

(21) In some embodiments, the materials of the bonding portions **112** include copper, nickel, titanium, cobalt, palladium, titanium, tin, silver, and/or alloys thereof or combinations thereof. In some embodiments, the materials of the conducting bumps **113** are the same as the materials of the dummy bumps **114**. In some embodiments, the materials of the conducting bumps **113** are different from the materials of the dummy bumps **114**. In some embodiments, the structure of each single conducting bump **113** is the same as the structure of each single dummy bump **114**. In some embodiments, the structure of at least one dummy bump **114** is different from the structures of the conducting bumps **113**. In some embodiments, the structures of the dummy bumps **114** are different from the structures of the conducting bumps **113**.

(22) As the dimension and the pitch of the bumps are continually scaled down, it is important to maintain good uniformity and coplanarity of the formed bumps in the bump array. By forming the dummy bumps surrounding the conducting bumps in the same process, the uniformity and coplanarity of the formed bumps are significantly improved especially for bumps in the peripheral region of the array.

(23) FIG. **11A** to FIG. **11D** are schematic cross-sectional views illustrating the bump structures of according to some embodiments of the present disclosure. FIG. **12A** to FIG. **12C** are schematic enlarged partial cross-sectional views showing the dummy bump attached to a portion of a package structure according to some exemplary embodiments of the present disclosure.

(24) Referring to FIG. **11A**, in some embodiments, the bump structure includes a metallic post **1100** and a stack of a first metal layer **1102**, a second metal layer **1104** and a solder material **1106** stacked sequentially on the metallic post **1100**. Referring to FIG. **11B**, in some embodiments, the bump structure includes a metallic post **1100** and a stack of a first metal layer **1102** and a solder material **1106** stacked sequentially on the metallic post **1100**. Referring to FIG. **11C** and FIG. **11D**, in some embodiments, the bump structure includes a metallic post **1100** and a first metal layer **1102** thereon, or the bump structure simply includes a metallic post **110**. In some embodiments, a material of the metallic post **1100** includes copper or a copper alloy. In some embodiments, a material of the first metallic layer **1102** includes nickel or a nickel alloy. In some embodiments, a material of the second metallic layer **1104** includes copper or a copper alloy. In some embodiments, a material of the solder material **1106** includes tin, copper, silver, bismuth, indium, zinc, antimony, manganese and/or alloys thereof. As described in the embodiments, the bonding portions **112** may include conducting bumps **113** and dummy bumps **114**, and the structures of the bonding portions **112** may include the bumps structures as illustrated in FIGS. **11A-11D**. In some embodiments, the conducting bumps **113** are formed with the bump structures as shown in FIG. **11A** or FIG. **11B**, while the dummy bumps **114** are formed with the bump structures as shown in FIG. **11C** or FIG.

11D. For example, the conducting bumps **113** and some or all of the dummy bumps **114** of the bonding portions **112** are formed with metallic posts at the same time through the same electroplating process, and then the conducting bumps **113** are later formed with solder caps to complete the bump structures.

(25) In some embodiments, referring to FIG. **12A**, the dummy bump **114** is located directly on the dielectric layer **1211** of the redistribution layer **1210**. The redistribution layer **1210** and the dielectric layer **1211** are similar to or substantially the same as the redistribution **110** and its topmost dielectric layer **106e** described in FIG. **1**. In some embodiments, referring to FIG. **12B**, the dummy bump **114** is located directly on the topmost dielectric layer **1211** of the redistribution layer **1210**, and covers the crater CT of the dielectric layer **1211**. In some embodiments, referring to FIG. **12C**, the dummy bump **114** covers the crater CT of the dielectric layer **1211** of the redistribution layer **1210** and the metallic pad P2 exposed by the crater CT, so that the dummy bump **114** is located directly on the electrically floating metallic pad P2. In the embodiments, the dummy bumps **114** as shown in FIGS. **12A-12C** are electrically floating.

(26) Referring to FIG. **2** and FIG. **3**, in some embodiments, another carrier **103** is provided and attached to the structure of the redistribution layer **110** with the bonding portions **112**, and the carrier **102** is detached through the debonding layer **104** and both are removed. Then, the whole structure JS is flipped so that the bottommost dielectric layer **106a** of the redistribution layer **110** is exposed and facing upward and portions of the bottommost conductive layer **108a** are exposed from the surface of the dielectric layer **106a**. In some embodiments, the bonding portions **112** are attached to the carrier **103**. In some embodiments, the carrier **103** is a carrying tape film.

(27) In some embodiments, referring to FIG. **3**, joining portions **118** are formed on the exposed portions (e.g. vias) of the conductive layer **108a** of the redistribution layer **110**. In some embodiments, the joining portions **118** are electrically connected with the redistribution layer **110** (e.g. the conductive layers **108a**, **108b**, **108c**, **108d**, **108e**). In embodiments, the formation of joining portions **118** includes forming a mask pattern (not shown) on the redistribution layer **110** with openings, forming a metallic material filling up the openings to form metallic portions and then removing the mask pattern.

(28) In some embodiments, the materials of the joining portions **118** may be selected from copper, cobalt, nickel, aluminum, tungsten, alloys or combinations thereof. In some embodiments, the joining portions **118** further optionally include an adhesion layer, a seed layer, pre-solder, solder paste and/or under-ball metallurgy (UBM) patterns formed on the surfaces of the joining portions for enhancing bonding. For example, the joining portions **118** may be formed by electroplating or deposition. In some embodiments, the conductive joining portions **118** are, for example, micro-bumps, metal posts, metal posts with solder paste, electroless nickel electroless palladium immersion gold (ENEPIG) formed bumps, or controlled collapse chip connection (C4) bumps.

(29) As seen in FIG. **4**, in some embodiments, at least one circuit substrate **200** is mounted and bonded to the structure JS. In some embodiments, the circuit substrate **200** includes a build-up board, a printed circuit board, a laminated board or a flexible laminate board. In some embodiments, the circuit substrate **200** may include one or more active components, passive components, or a combination thereof. In some embodiments, the circuit substrate **200** includes a dielectric material core structure **201** sandwiched between dielectric material layers **203**, and insulting layers **205** laminated on the dielectric material layers **203**, as well as metallization patterns including pads **206** embedded in the dielectric material layers **203**. In some embodiments, the metallization patterns are designed to electrically connect the various components such as the active components and/or passive components embedded in the circuit substrate to form functional circuitry. In some embodiments, the circuit substrate **200** is provided with joining connectors **210** attached to the pads **206** of the circuit substrate **200**. The circuit substrate **200** may provide single-side or dual-side electrical connection.

(30) Referring to FIG. **4**, the circuit substrate **200** is mounted to the structure JS, and the joining

portions **118** of the structure JS are respectively bonded with the joining connectors **210** located on the pads **206** of the circuit substrate **200**. In some embodiments, through performing a reflow process, the joining portions **118** of the structure JS are joined and fused with the joining connectors **210** on the pads **206** of the substrate **200**. In some embodiments, the reflow process includes performing a thermal compression bonding process to turn the joining portions **118** into a melted state or a semi-melted state to integrate and bond with the connectors **210** located between the structure JS and the circuit substrate **200**. The reflow temperature may be higher than a melting point of joining portions **118** and/or the solder paste.

(31) In FIG. **4**, in some embodiments, an underfill **240** is formed and filled between the structure JS and the circuit substrate **200**. In some embodiments, the underfill **240** is filled between the structure JS and the circuit substrate **200** and filled between the fused connectors **210** that are located between the structure JS and the circuit substrate **200**. In some embodiments, the underfill **240** filled between the structure JS and the circuit substrate **200** can protect the fused connectors **210** against thermal or physical stresses and further secure the bonding of the structure JS with the circuit substrate **200**.

(32) In some embodiments, the underfill **240** is formed by capillary underfill filling (CUF) and the underfill **240** not only fills between the redistribution layer **110** and the circuit substrate **200** but also fills up the gaps between the connectors **210**. In some embodiments, a curing process may be performed to solidify the underfill **240**.

(33) In some embodiments, referring to FIG. **5**, after the underfill **240** is formed, a dicing process is performed to the structure JS along the scribing lanes SL (cutting lanes). In some embodiments, the dicing process is performed to cut the whole structure JS (at least cutting through the redistribution layer **110**) and the bonded circuit substrate **200** into individual and separated units **11**. In some embodiments, the dicing process is performed to cut through the structure JS along the scribing lanes SL without cutting into the corresponding circuit substrate **200** and the underfill **240**. Later, the individual unit **11** is flipped and transferred to another carrier **105**. In some embodiments, the carrier **105** is a carrying tape film. In one embodiment, the dicing process is a wafer dicing process including mechanical sawing, blade dicing and/or laser cutting. In some embodiments, as the dummy bumps **114** are located beside the scribing lanes SL, the cutting blade or dicing saw cutting through the redistribution layer **110** does not cut into or pass through the dummy bumps **114**. That is, the dummy bumps are not diced or damaged during the dicing process.

(34) Referring back to FIG. **6**, at least one first semiconductor die **120** and two or more second semiconductor dies **130** are provided and placed on the redistribution layer **110** over the carrier **105**. In some embodiments, the first semiconductor die **120** and the second semiconductor die(s) **130** are disposed in the die attaching region DA and attached onto the top surface **110T** of the redistribution layer **110**. In FIG. **6**, three dies are shown as the exemplary dies of the package structure, but it is understood that multiple dies or two or more types of dies or different types of dies may be included within the package structure. In some embodiments, the first semiconductor die **120** and the second semiconductor die(s) **130** are different types of dies or perform different functions. In some embodiments, the first semiconductor die **120** may include one or more of an application-specific integrated circuit (ASIC) chip, an analog chip, a sensor chip, a wireless application chip (such as a Bluetooth chip or a radio frequency chip), a voltage regulator chip or a system-on-a chip (SoC). In some embodiments, the second semiconductor dies **130** include one or more memory chips, such as high bandwidth memory (HBM) chips, dynamic random access memory (DRAM) chips or static random access memory (SRAM) chips. In some embodiments, one of the second dies **130** may be a memory die including memory chips, and the other die **120** may be a SoC die including a controller chip. In certain embodiments, dies and chips may be used interchangeably.

(35) In certain embodiments, in FIG. **6**, the first and second semiconductor dies **120**, **130** are respectively provided with contacts **122**, **132** facing downward, after aligning the contacts **122**, **132**

with the corresponding conducting bumps **113**, the contacts **122**, **132** of the semiconductor dies **120**, **130** are respectively bonded to the conducting bumps **113** of the bonding portions **112** on the redistribution layer **110**. In one embodiment, the bonding of the semiconductor dies **120**, **130** to the redistribution layer **110** includes performing a reflow process or a thermal compression bonding process to bond the contacts **122**, **132** with the conducting bumps **113** to become fused connectors **115**. In some embodiments, through the conducting bumps **113** and the contacts **122**, **132** (i.e. through the fused connectors), the first and second semiconductor dies **120**, **130** are electrically connected with the redistribution layer **110**, while the dummy bumps are electrically unconnected with the first and second semiconductor dies **120**, **130**. In some embodiments, the dummy bumps **114** are electrically floating. The dummy bumps **114** are not electrically connected or coupled with any of the semiconductor dies. In some embodiments, the contacts **122**, **132** are micro-bumps, metallic posts such as copper posts, copper alloy posts or other suitable metallic connectors. In certain embodiments, the semiconductor dies **120**, **130** are mounted onto the redistribution layer **110** over the carrier **102** side-by-side, and the number of the dies arranged side-by-side or stacked over another die(s) may be adjusted or modified based on the product design but are not limited by the exemplary embodiments. In some embodiments, as illustrated in FIG. 5 and FIG. 6, based on the product design and the locations of the scribe lanes SL, at least one first semiconductor die **120** and multiple second semiconductor die **130** are included in each package unit **11** after dicing.

(36) As seen in FIG. 6, after mounting and bonding the first and second semiconductor dies **120**, **130** to the redistribution layer **110**, some of the dummy bumps **114** are located right below the second semiconductor dies **130** (within the die attaching region DA), and are not connected with the contacts of the above semiconductor dies. In some embodiments, the dummy bumps **114** that are located right below the second semiconductor dies **130** are physically separate and spaced apart from the above second semiconductor dies **130**. Although the dummy bumps **114** are shown in FIG. 6 as located below the second semiconductor dies **130**, it is possible that the dummy bumps are located below other different semiconductor dies or both, depending on the layout arrangement of the dies.

(37) According to the above exemplary embodiments, the layout and configuration of the redistribution layer **110** may be suitably formed within the wafer-level package structures. Although one frontside redistribution layer is described in the above embodiments, more than one or multiple redistribution layers (RDLs) may be provided in the package structure or arranged on both front side and back side of the die(s) or chip(s) for signal redistributions among multiple dies or chips. Additionally, the semiconductor package **11** may further include additional dies or sub-package units disposed over or below the dies and another redistribution layer(s) may be formed to electrically connect the additional dies or sub-package units. The structures and/or the processes of the present disclosure are not limited by the exemplary embodiments. In some embodiments, the sizes or dimensions of the connectors **210** are larger than the sizes or dimensions of the fused connectors **115** in the package structure **11**.

(38) Referring to FIG. 7, an underfill **136** is filled between the first and second semiconductor dies **120**, **130** and the redistribution layer **110**, encapsulating the fused connectors **115** and covering the dummy bumps **114**.

(39) FIG. 9 illustrates an exemplary schematic top view of a package unit of the package structure. In some embodiments, as seen in the schematic top view, four second semiconductor dies **130** are arranged at two opposite sides of the first semiconductor die **120**. In some embodiments, the first semiconductor die **120** has a larger die size and/or has multiple functionality, while the second semiconductor dies **130** have a smaller die size. From the planar top view of FIG. 9, considering the semiconductor dies **120**, **130** being transparent, the locations of the semiconductor dies **120**, **130** are denoted by the dashed lines, the distribution regions (active regions) AA of the conducting bumps **113** in the semiconductor dies **120**, **130** are shown as shaded regions, and the borders of the die attaching region DA (defined by the outer sidewalls of the dies **120**, **130** and labeled as the bold

dotted line in FIG. 9) are vertically aligned with the outer sidewalls of the dies. As seen in FIG. 6 and from the planar top view of FIG. 9, the distribution span DD of the dummy bumps 114 is partially overlapped with the die attaching region DA, and the dummy bumps 114 are arranged around the periphery or the rim of the die attaching region DA. For example, considering the distribution region AA of the conducting bumps 113 of the semiconductor die 120 has a square shape with a distribution distance DS1, the distribution distance DS2 of the distribution span DD of the dummy bumps 113 ranges from about 0.01% to about 30% of the distance DS1.

(40) In some embodiments, using round shaped bumps as examples, the size (diameter) of the conducting bumps 113 is substantially the same as the size (diameter) of the dummy bumps 114. From the planar top view of FIG. 9, the dummy bumps 114 are arranged along outward sides (non-facing sides) of the first and second semiconductor dies 120, 130 (i.e. along the rim of the die attaching region DA) and surrounding the die attaching region DA.

(41) In some embodiments, as seen in FIG. 10, some dummy bumps 114' are arranged around the facing sides between the two adjacent second semiconductor dies 130, and the dummy bumps 114' are of a smaller size than that of the dummy bumps 114. However, no dummy bumps are located between the two facing sides of the adjacent first and second semiconductor dies 120, 130. From the planar top view of FIG. 10, except for the dummy bumps 114 arranged as two rows around the rim of the die attaching region DA, smaller dummy bumps 114' are arranged around the right side of the die attaching region DA (i.e. around the outward sides of the two second semiconductor dies 130 located at the right side). In some embodiments, using round shaped bumps as examples, the size (diameter) of the conducting bumps 113 is larger than the size (diameter) of the dummy bumps 114.

(42) It is understood that the number of the rows of the dummy bumps are not limited to the exemplary drawings herein. Depending on the product requirements, the rows of the dummy bumps may range from two rows to thirty rows or even to sixty rows. In some embodiments, by arranging ten rows to thirty rows of dummy bumps surrounding the outer sides of the adjacent semiconductor dies, the coplanarity of the overall formed bumps (i.e. the total thickness variation) is improved by about 12% to about 36%. In some embodiments, the arrangement of the dummy bumps may be symmetric or asymmetric, relative to the layout of the dies. Also, two or more types of dummy bumps may be formed, with different structures, dimensions and/or materials, depending on the layout design.

(43) Referring back to FIG. 7, the underfill 136 filled between the first and second semiconductor dies 120, 130 and the redistribution layer 110 encapsulates the fused connectors 115 for better attachment. In some embodiments, the underfill 136 also encapsulates the dummy bumps 114. Also, by filling the gaps between the first and second semiconductor dies 120, 130, the redistribution layer 110, and the fused connectors 115, the underfill 136 relieves the thermal stress concentrated on the joints of the connectors and the joint reliability is enhanced. Due to the existence of the dummy bumps 114 and 114', the underfill 136 has a steep profile as the extension range of the underfill 136 is guided by the arrangement and distribution of the dummy bumps. Compared with the structure without the dummy bumps and with overflowing underfill, the underfill 136 formed in the structure with the dummy bumps 114 and 114' has an average extension distance about 10%-40% shorter, or at least 25% shorter. That means, for the structure with dummy bumps, the underfill 136 spreads out to cover mainly the dummy bumps 114 with a shorter extension distance. In general, the dummy bumps guide and constrain the extension range of the underfill. In some embodiments, the underfill 136 has an average extension distance, which is obtained from averaging the extension distances D1 in the X-direction or Y-direction (the shortest distance measuring from the sidewall of the die(s) to the edge of the underfill 136), ranging from 50 microns to 5000 microns. In some embodiments, the underfill 136 has a height Z1 (measuring from the top surface 110T of the redistribution layer 110 to the topmost point of the underfill 136 in the thickness direction Z) substantially about the same as the height of the second semiconductor

die(s) **130**. Through the arrangement of the dummy bumps, the underfill is formed with a steep profile, and the underfill not only strengthens the bonding and attachment of the dies with the redistribution layer but also provides higher layout flexibility for being space economical. In some embodiments, as the underfill **136** is constrained by the locations of the dummy bumps **114**, the formed underfill **136** does not overflow outside the package unit and does not cover the sidewalls of the cut redistribution layer **110** of the package structure **11**.

(44) In some embodiments, as seen in FIG. 7, the underfill **136** substantially fills the gaps between the fused connectors **115**, and fills between the first and second semiconductor dies **120** and **130**. In some embodiments, the underfill **136** is formed as a void-free filling material filling up the spaces between dies and the redistribution layer **110** and between the fused connectors **115**. In some embodiments, the material of the underfill **136** includes epoxy resins or other suitable polymer material and optionally fillers such as silica or alumina. In some embodiments, the underfill **136** is formed by performing a capillary filling process and then a low temperature curing process. In some embodiments, the underfill **136** not only fills up the gaps between dies and the redistribution layer **110** and between the fused connectors **115** but also spreads over to cover the sidewalls **130S** of the second semiconductor dies **130**. In some embodiments, the sidewalls **130S** of the second semiconductor dies **130** are fully covers by the underfill **136**. In some embodiments, the underfill **136** encapsulates the dummy bumps **114** and extends from sidewalls **130S** of the respective second semiconductor dies **130** outwardly until it reaches the outmost dummy bumps **114** (e.g. the outmost edges of the dummy bumps **114**) of the outmost row. That means that the underfill **136** extends horizontally from outward sidewalls of the dies **130** to reach and fully cover the dummy bumps **114**. As the dummy bumps **114** are spaced apart and unconnected with the respective second semiconductor dies **130**, the underfill **136** is the only thing sandwiched between the dummy bumps **114** and the respective second semiconductor dies **130**. In some embodiments, the underfill **136** is in direct physical contact with the dummy bumps **114** and wraps around the dummy bumps **114** that are located under the second semiconductor dies **130** and outside the second semiconductor dies **130** on the redistribution layer **110**. From the schematic top view of FIG. 9, some of the dummy bumps **114** that are located below the second semiconductor dies **130** fall within the vertical projections (the dashed lines) of the second semiconductor dies **130**, while some of the dummy bumps **114** that are located beside the second semiconductor dies **130** are located outside the vertical projections (the dashed lines) of the second semiconductor dies **130**.

(45) Through these conductive connections and the redistribution structure, the semiconductor dies **120**, **130** of finer pitches are electrically connected with the circuit substrate **200** of further larger pitches.

(46) FIG. 8 is a schematic cross-sectional view of a package structure with dummy bumps according to an exemplary embodiment of the present disclosure.

(47) As seen in FIG. 8, in some embodiments, a package structure **18** including a package unit **12** mounted and bonded to a circuit substrate **200** is illustrated. In some embodiments, the package unit **12** includes dies **120**, **130** bonded to the redistribution layer **110**, dummy bumps **114** located on the redistribution layer **110** and an underfill **136** filled between the dies **120**, **130** and the redistribution layer **110**. In some embodiments, the dies **120**, **130**, the redistribution layer **110**, the dummy bumps **114**, the underfill **136** and the circuit substrate **200** are substantially the same or similar to the corresponding elements as described in the above paragraphs, and similar or the same elements may be denoted using the same reference labels in various embodiments. As seen in FIG. 8, the circuit substrate **200** may provide single-side or dual-side electrical connection and prove further electrical connection through the conductive balls **280**.

(48) Referring to FIG. 8, the package structure **12** further includes a molding compound **140** on the redistribution layer **110** encapsulating the semiconductor dies **120**, **130** and the underfill **136**. In some embodiments, the molding compound **140** at least laterally wraps around the underfill **136** as well as the semiconductor dies **120**, **130** on the redistribution layer **110**. In one embodiment, the

material of the molding compound **140** includes epoxy resins, phenolic resins or silicon-containing resins. In some embodiments, the material of the molding compound **140** includes filler particles such as silica particles. In some embodiments, the molding compound **140** may be over-molded to be higher than the dies **120**, **130**. In some embodiments, the molding compound **140** is over-molded and then planarized to further reduce the thickness of the molding compound **140**.

(49) In FIG. **8**, in some embodiments, an underfill **240** is formed and filled between the package structure **12** and the circuit substrate **200**. In some embodiments, the underfill **240** not only fills up the gaps between the package structure **12** and the circuit substrate **200** but also overflows to partially cover the sidewalls of the package **12** (covers the sidewalls of the diced redistribution layer **110**).

(50) According to the above exemplary embodiment, the package structure(s) may be suitably formed following the processes for fabricating the integrated fan-out (InFO) wafer-level package structure. More than one or multiple redistribution layers (RDLs) may be provided in the package structure or arranged on both front side and back side of the die(s) or chip(s) for signal redistributions among multiple dies or chips. The structures and/or the processes of the present disclosure are not limited by the exemplary embodiments.

(51) FIG. **13** and FIG. **14** are schematic cross-sectional views of package structures with dummy bumps according to some exemplary embodiments of the present disclosure.

(52) In some embodiments, referring to FIG. **13**, the semiconductor package **13** includes an organic substrate **300**, conductive connectors **380** disposed on the bottom side of the organic substrate **300**, and a multilayered structure **400** disposed on the top side of the organic substrate **300** and connected with the organic substrate **300** through conductive connectors **370**. In some embodiments, the semiconductor package **13** includes a first integrated circuit (IC) module **310**, a second IC module **320** and a third IC module **330** disposed on the multilayered structure **400**. In some embodiments, the first, second and third IC modules **310**, **320**, **330** are different types of modules and have different functionalities. For example, either of the first, second and third IC modules **310**, **320**, **330** may include one or two or more types of semiconductor dies, such as a logic die (e.g., central processing unit (CPU), graphics processing unit (GPU), system-on-a-chip (SoC), application processor (AP), microcontroller, etc.), a memory die (e.g., dynamic random access memory (DRAM) die, static random access memory (SRAM) die, a hybrid memory cube (HMC), a high bandwidth memory (HBM) die, etc.), a power management die (e.g., power management integrated circuit (PMIC) die), a radio frequency (RF) die, a sensor die, a micro-electro-mechanical-system (MEMS) die, a signal processing die (e.g., digital signal processing (DSP) die), a front-end die (e.g., analog front-end (AFE) dies), the like, or combinations thereof. In some embodiments, the first IC module **310** includes a logic die, the second IC module **320** includes a power management die and the third IC module **330** includes a memory die. In one embodiment, the IC modules may include the package(s) similar or the same configurations as described above, and the multilayered structure **400** may include the redistribution layer(s) as described above, and the similar structures may be fabricated following the manufacturing method as depicted from FIG. **1** to FIG. **8**. In some embodiments, the multilayered structure **400** provides electrical pathing and connection between the IC modules and the organic substrate **300** by way of conductive bumps **370**. In some embodiments, the organic substrate **300** provides structural support for the package, as well as providing electrical signal routing between the IC modules and the laminate circuits such as printed circuit board (PCB) or other sub-packages, but not limited thereto.

(53) In some embodiments, the first, second and third IC modules **310**, **320**, **330** are bonded to the multilayered structure **400** through connectors **350**, and the connectors **350** electrically connect the first, second and third IC modules **310**, **320**, **330** with the multilayered structure **400** and the organic substrate **300**. As seen in FIG. **13**, the semiconductor package **13** also includes dummy bumps **360** located between the first, second and third IC modules **310**, **320**, **330** and the top surface **400T** of the multilayered structure **400** and located on the top surface **400T** of the

multilayered structure **400**, and the dummy bumps **360** are electrically floating. Referring to FIG. **13**, for the first, second and third IC modules **310**, **320**, **330**, the dummy bumps **360** are located under (not in contact with) the above modules, and are arranged as outer rows surrounding the connectors **350**. Referring to FIG. **13**, for the first, second and third IC modules **310**, **320**, **330**, underfills **366A**, **366B**, **366C** are respectively formed between the first, second and third IC modules **310**, **320**, **330** and the multilayered structure **400**, and the underfills **366A**, **366B**, **366C** are respectively filled between the corresponding first, second and third IC modules **310**, **320**, **330** and the top surface **400T** of the multilayered structure **400** and cover the top surface **400T** of the multilayered structure **400**. In some embodiments, the underfills **366A**, **366B**, **366C** fill the gaps between the connectors **350** located between the first, second and third IC modules **310**, **320**, **330** and the top surface **400T** of the multilayered structure **400**.

(54) In some embodiments, as seen in FIG. **13**, the underfill **366A** encloses the dummy bumps **360** and fully covers the sidewalls **310S** of the first IC module **310** (extending from the top side of the sidewalls **310S** to the top surface **400T** of the multilayered structure **400**). In some embodiments, as seen in FIG. **13**, the underfill **366B** encloses the dummy bumps **360** and fully covers the sidewalls **320S** of the second IC module **320**. In some embodiments, as seen in FIG. **13**, the underfill **366C** encloses the dummy bumps **360** and partially covers the sidewalls **330S** of the third IC module **330** (extending from the upper parts of the sidewalls **330S** to the top surface **400T** of the multilayered structure **400**). As seen in at the upper left part of FIG. **13**, the relative layout of the modules and the distribution regions of the dummy bumps are shown, the first, second and third IC modules **310**, **320**, **330** (vertical projections shown in dotted lines) are arranged side by side and spaced apart from one another, the distribution regions **R310**, **R320**, **R330** of the dummy bumps **360** that are disposed under the corresponding first, second and third IC modules **310**, **320**, **330** are shown as three separate rings surrounding the corresponding first, second and third IC modules **310**, **320**, **330** and partially overlap with the vertical projections of the corresponding first, second and third IC modules **310**, **320**, **330**.

(55) Referring to FIG. **14**, in some embodiments, the dummy bumps **360**, for the first, second and third IC modules **310**, **320**, **330**, are located under (not in contact with) the above modules, and are arranged as rows surrounding the outer sidewalls of the modules. Referring to FIG. **14**, for the first IC module **310**, an underfill **366A** is filled between the first IC module **310** and the multilayered structure **400**, and the underfill **366A** encloses the dummy bumps **360** and fully covers the sidewalls **310S** of the first IC module **310** (extending from the top side of the sidewalls **310S** to the top surface **400T** of the multilayered structure **400**). In FIG. **14**, in some embodiments, the second and third IC modules **320**, **330** are disposed as one group, the dummy bumps **360** are arranged around the outward sidewalls of the second and third IC modules **320**, **330** and no dummy bumps are arranged along the facing sidewalls of the second and third IC modules **320**, **330**. In some embodiments, as seen in FIG. **14**, the underfill **366B** encloses the dummy bumps **360** and fully covers the outward sidewalls **320S**, **330S** of the IC modules **320**, **330**. As seen in at the upper left part of FIG. **14**, the relative layout of the modules and the distribution regions of the dummy bumps are shown, the first, second and third IC modules **310**, **320**, **330** (vertical projections shown in dotted lines) are arranged side by side and spaced apart from one another, the distribution region **R310** of the dummy bumps **360** under and around the first IC module **310** is shown as a ring and partially overlap with the vertical projections of the first IC module **310**, while the distribution region **R325** of the dummy bumps **360** that are disposed under and around the second and third IC modules **320**, **330** is shown as another individual ring surrounding the corresponding second and third IC modules **320**, **330** and partially overlap with the vertical projections of the second and third IC modules **320**, **330**.

(56) Due to the existence of the dummy bumps, the underfill(s) has a steep profile with a short extension distance and almost covers the whole sidewalls of the module(s). Through the arrangement of the dummy bumps, the steep profiled underfill strengthens the bonding and

attachment of the connectors and less overflow of the underfill leads to layout flexibility and larger process windows. Through the formation of the dummy bumps, the formed bumps and connectors have better uniformity and less thickness variation, leading to satisfactory bonding reliability for the package structure.

(57) The disclosure is not limited neither by the type nor the number of semiconductor packages connected to the circuit substrate. It will be apparent that different types of semiconductor package units may be used to produce semiconductor device package structures including the circuit substrate disclosed herein, and all these semiconductor devices are intended to fall within the scope of the present description and of the attached claims. For example, Chip-On-Wafer-On-Substrate (CoWoS) structures, three-dimensional integrated circuit (3DIC) structures, Chip-on-Wafer (CoW) packages, Package-on-Package (PoP) structures may all be used as the semiconductor package units, alone or in combination.

(58) According to some embodiments, a semiconductor package has a redistribution layer, a semiconductor die, conducting connectors, dummy bumps and an underfill. The semiconductor die is disposed on a top surface of the redistribution layer and electrically connected with the redistribution layer. The conducting connectors are disposed between the semiconductor die and the redistribution layer, and are physically and electrically connected with the semiconductor die and the redistribution layer. The dummy bumps are disposed on the top surface of the redistribution layer, beside the conducting connectors and under the semiconductor die. The underfill is disposed between the semiconductor die and the redistribution layer and sandwiched between the dummy bumps and the semiconductor die. The dummy bumps are electrically floating. The dummy bumps are in contact with the underfill without contacting the semiconductor die.

(59) According to some embodiments, a package structure includes at least a first die, a second die, a redistribution layer, dummy bumps, an underfill and a circuit substrate. The first die and the second die are arranged side by side. The circuit substrate is disposed under the first and second dies and electrically connected with the first die and second die. The redistribution layer is disposed between the first and second dies and the circuit substrate. The redistribution layer has a die attaching region, the first die and second die are disposed on a first side of the redistribution layer and within the die attaching region. The redistribution layer is electrically connected with the first die and the second die through connectors therebetween. The dummy bumps are disposed on the first side of the redistribution layer and spaced apart from the first die and the second die. The dummy bumps are electrically unconnected with the first die and the second die. The underfill is disposed between the first die, the second die and the redistribution layer and filled between the dummy bumps and the first die and the second die. The dummy bumps are distributed over a distribution span and the distribution span is partially overlapped with the die attaching region.

(60) According to some embodiments, a manufacturing method for a package structure is provided. A redistribution layer having dielectric layer and conductive layers stacked in alternation is formed. Conducting bumps and dummy bumps are formed on a top surface of the redistribution layer. A semiconductor die with joining portions is provided. The semiconductor die is disposed onto the redistribution layer, and the joining portions are aligned with the conducting bumps on the top surface of the redistribution layer. The semiconductor die is bonded to the redistribution layer by joining the conducting bumps with the joining portions, so that the semiconductor die is electrically connected to the redistribution layer. An underfill is dispensed between the semiconductor die and the redistribution layer and between the semiconductor die and the dummy bumps.

(61) The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that

they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A semiconductor package comprising: a redistribution layer; a semiconductor die, disposed on a top surface of the redistribution layer and electrically connected with the redistribution layer; conducting connectors, disposed between the semiconductor die and the redistribution layer, and physically and electrically connected with the semiconductor die and the redistribution layer; dummy bumps, disposed on the top surface of the redistribution layer, beside the conducting connectors and under the semiconductor die; and an underfill, disposed between the semiconductor die and the redistribution layer, wrapping around the conducting connectors and extending between the dummy bumps and sidewalls of the semiconductor die, wherein the underfill fully covers outermost sidewalls of the semiconductor die without covering a top surface of the semiconductor die, the dummy bumps are electrically floating and the dummy bumps are in contact with the underfill without contacting the semiconductor die, and wherein second dummy bumps of the dummy bumps are located outside a vertical projection of the semiconductor die, and the underfill extends outwardly from the outermost sidewalls of the semiconductor die to outermost second dummy bumps of the second dummy bumps.
2. The semiconductor package as claimed in claim 1, wherein the dummy bumps are distributed over a distribution span, and the distribution span is partially overlapped with the vertical projection of the semiconductor die on the top surface of the redistribution layer.
3. The semiconductor package as claimed in claim 2, wherein the dummy bumps are arranged surrounding the conducting connectors and the distribution span of the dummy bumps is in a ring shaped from a top view.
4. The semiconductor package as claimed in claim 2, wherein first dummy bumps of the dummy bumps are located directly under the semiconductor die and fall within the vertical projection of the semiconductor die, the second dummy bumps of the dummy bumps are located along a border of the vertical projection and the first dummy bumps and the second dummy bumps are located at a same side of the redistribution layer.
5. The semiconductor package as claimed in claim 1, wherein the dummy bumps are located directly on a dielectric layer of the redistribution layer.
6. The semiconductor package as claimed in claim 1, wherein the underfill surrounds the conducting connectors.
7. The semiconductor package as claimed in claim 1, wherein the dummy bumps include micro bumps.
8. The semiconductor package as claimed in claim 1, wherein the dummy bumps include metallic posts.
9. A package structure comprising: a first die and a second die, arranged side by side, wherein the first die has a first height and the second die has a second height smaller than the first height; a circuit substrate, disposed under the first and second dies and electrically connected with the first die and second die; a redistribution layer, disposed between the first and second dies and the circuit substrate, wherein the redistribution layer has a die attaching region, the first die and second die are disposed on a first side of the redistribution layer and within the die attaching region, and the redistribution layer is electrically connected with the first die and the second die through connectors therebetween; dummy bumps, disposed on the first side of the redistribution layer and spaced apart from the first die and the second die, where dummy bumps are electrically unconnected with the first die and the second die; and an underfill, disposed between the first die, the second die and the redistribution layer and filled between a front side surface of the redistribution layer, the dummy bumps and the first die and the second die; and wherein the dummy

bumps are distributed over a distribution span and the distribution span is partially overlapped with the die attaching region, and the underfill extending from outer sidewalls of the first die to the front side surface of the redistribution layer has a third height larger than the second height and about the same with the first height, and wherein the underfill extends outwardly from the outer sidewalls of the first die and outer sidewalls of the second die to reach and cover the dummy bumps that are located outside the die attaching region.

10. The structure as claimed in claim 9, wherein the dummy bumps that are located directly under the first die or the second die are located in the die attaching region, and the dummy bumps that are located outside the die attaching region are arranged along a periphery of the die attaching region.

11. The structure as claimed in claim 9, wherein the dummy bumps are arranged surrounding the connectors and the distribution span of the dummy bumps is in a ring shaped from a top view.

12. The structure as claimed in claim 9, wherein the underfill fills between the connectors.

13. The structure as claimed in claim 9, wherein the redistribution layer includes multiple dielectric layers and multiple conductive layers arranged in alternation, and the dummy bumps are disposed directly on a topmost dielectric layer of the multiple dielectric layers of the redistribution layer.

14. The structure as claimed in claim 13, wherein the connectors are disposed on a topmost conductive layer of the multiple conductive layers of the redistribution layer.

15. The structure as claimed in claim 9, wherein dummy bumps are electrically floating and electrically unconnected with the redistribution layer, the first die and the second die.

16. The structure as claimed in claim 9, further comprising conductive connectors located between the circuit substrate and the redistribution layer.

17. The structure as claimed in claim 12, wherein the underfill covers the whole outer sidewalls of the first die and the second die without covering sidewalls of the redistribution layer and top surfaces of the first and second dies.

18. A package structure comprising: first dies and a second die, arranged side by side and spaced apart from one another, wherein the second die located between the first dies has a second height smaller than a first height of the first dies; a circuit substrate, disposed under the first and second dies and electrically connected with the first die and second die; a redistribution layer, disposed between the first and second dies and the circuit substrate, wherein the first dies and second die are disposed on a first side of the redistribution layer, and the redistribution layer is electrically connected with the first dies and the second die through connectors therebetween; dummy bumps, disposed on the first side of the redistribution layer and spaced apart from the first dies and the second die, wherein the dummy bumps are electrically floating and unconnected with the first dies and the second die; and an underfill, disposed between the first dies, the second die and the redistribution layer and filled between a front side surface of the redistribution layer, the dummy bumps and the first dies and the second die; and wherein the underfill fully covers outer sidewalls of the first dies and outer sidewalls of the second die, and an extension height of the underfill measuring from the front side surface of the redistribution layer to a topmost point of the underfill is larger than the second height and about the same with the first height, and wherein the underfill extends outwardly from the outer sidewalls of the first dies and the second die to reach and cover the dummy bumps that are located outside vertical projections of the first dies and the second die.

19. The structure as claimed in claim 18, wherein dummy bumps include first round-shaped bumps located directly under the first dies, and second round-shaped bumps located between the first dies, and the underfill fully covers the first and second round-shaped bumps.

20. The structure as claimed in claim 19, wherein the second round-shaped bumps are smaller in sizes than those of the first round-shaped bumps.
